

CS543 Web Services – Assignment 3

Integrate an External SOAP Partner

Project: CampusEats

Team ID = Group No. 10

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1. Context

CampusEats integrates with an external payment partner named **CampusPay Payment Gateway**. The single operation integrated in this assignment is **chargePayment**, which is used to request payment for a CampusEats order. SOAP is selected for this external payment edge because the partner contract is explicitly described by a strong WSDL contract, SOAP headers can carry message-level authentication information, and SOAP provides structured fault handling suitable for a critical payment operation. The remaining CampusEats services can continue using REST because they are internal application services and do not require this external SOAP contract.

2. HTTP Binding

The SOAP message is carried using HTTP POST. The endpoint and SOAPAction below match the WSDL service/port and binding.

```
POST /soap/payment HTTP/1.1
Host: api.campus-pay.example.com
Content-Type: text/xml; charset=utf-8
SOAPAction: "http://campus-pay.example.com/payment/chargePayment"
```

```
Endpoint:
https://api.campus-pay.example.com/soap/payment
```

3. Discovery

CampusEats would obtain the partner contract either from a direct WSDL URL supplied by the partner or from a modern service catalogue/registry lookup. No live UDDI server is required. The registry record below acts as a tModel-style pointer to the technical WSDL contract.

Business	CampusPay
Service	Payment Gateway SOAP Service
Endpoint	https://api.campus-pay.example.com/soap/payment
WSDL / tModel-style pointer	https://api.campus-pay.example.com/wSDL/payment.wsdl

4. Fault Mapping

If CampusPay returns the partner-specific fault **card_declined**, CampusEats translates it at the integration boundary into its own contract-level error **PAYMENT_DECLINED**. The partner vocabulary is therefore not exposed to students or CampusEats clients. A user-facing message can simply be: **Payment declined. Please try another payment method.**

5. Submitted SOAP Artifacts

The companion files contain the complete partner WSDL, the SOAP request with credentials in the Header and chargePayment call in the Body, a successful SOAP response, and a realistic SOAP Fault. The XML messages are written to match the operation, request/response elements, binding, and endpoint defined by the WSDL.